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Set-7

13. Write a Pandas program to detect missing values of a given DataFrame. Display True or False.

Aim

To write a Pandas program to detect missing values in a given DataFrame and display True for missing values and False for non-missing values.

Procedure

1. Import the Pandas library.
2. Create a DataFrame containing some missing (NaN) values.
3. Use the `isnull()` function to detect missing values.
4. Display the resulting DataFrame containing **True** and **False**.
5. **True** indicates a missing value, while **False** indicates a non-missing value.

The screenshot shows a Python IDE with two windows. The left window displays the code for creating a DataFrame with missing values and using the `isnull()` function to detect them. The right window shows the output of the program, including the DataFrame and the resulting boolean values for missing data.

```
exp13.py - C:/Users/sandr/exp13.py (3.12.10)
File Edit Format Run Options Window Help
import pandas as pd

# Create DataFrame
data = {
    'Name': ['Alice', 'Bob', None, 'David'],
    'Age': [20, None, 22, 21],
    'Marks': [85, 90, None, 88]
}

df = pd.DataFrame(data)

print("DataFrame:")
print(df)

print("\nMissing Values (True/False):")
print(df.isnull())
```

Python 3.12.10 [tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36] [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

```
>>>
===== RESTART: C:/Users/sandr/exp13.py =====
===== RESTART: C:/Users/sandr/exp13.py =====

DataFrame:
   Name  Age  Marks
0  Alice  20.0   85.0
1   Bob   NaN   90.0
2   NaN   22.0   NaN
3  David  21.0   88.0

Missing Values (True/False):
   Name  Age  Marks
0  False False False
1  False  True  False
2   True  False  True
3  False False  False

>>>
```

Result

Thus, the Pandas program was successfully executed to detect missing values in the DataFrame. **True** represents missing values and **False** represents non-missing values.

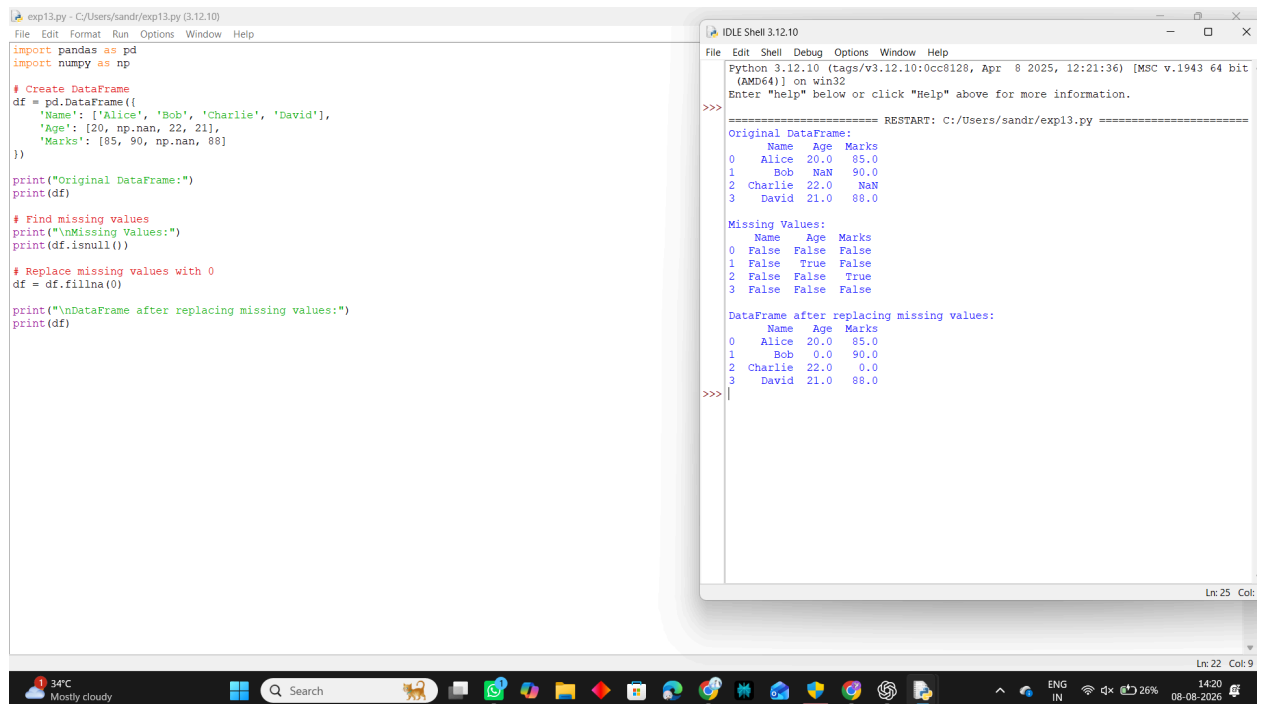
14. Write a Pandas program to find and replace the missing values in a given DataFrame which do not have any valuable information.

Aim

To write a Pandas program to find and replace missing values in a given DataFrame with suitable values when the missing data does not contain valuable information.

Procedure

1. Import the Pandas library.
2. Create a DataFrame containing missing (NaN) values.
3. Use `isnull()` to identify the missing values.
4. Use `fillna()` to replace the missing values.
5. Display the original and updated DataFrame.



The screenshot shows a Python IDE with two windows. The left window displays a Python script that creates a DataFrame with missing values, identifies them using `isnull()`, and replaces them with 0 using `fillna(0)`. The right window shows the output of the script, including the original DataFrame, the missing values, and the updated DataFrame.

```
exp13.py - C:/Users/sandr/exp13.py (3.12.10)
File Edit Format Run Options Window Help

import pandas as pd
import numpy as np

# Create DataFrame
df = pd.DataFrame({
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],
    'Age': [20, np.nan, 22, 21],
    'Marks': [85, 90, np.nan, 88]
})

print("Original DataFrame:")
print(df)

# Find missing values
print("\nMissing Values:")
print(df.isnull())

# Replace missing values with 0
df = df.fillna(0)

print("\nDataFrame after replacing missing values:")
print(df)
```

Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/sandr/exp13.py =====

Original DataFrame:

	Name	Age	Marks
0	Alice	20.0	85.0
1	Bob	NaN	90.0
2	Charlie	22.0	NaN
3	David	21.0	88.0

Missing Values:

	Name	Age	Marks
0	False	False	False
1	False	True	False
2	False	False	True
3	False	False	False

DataFrame after replacing missing values:

	Name	Age	Marks
0	Alice	20.0	85.0
1	Bob	0.0	90.0
2	Charlie	22.0	0.0
3	David	21.0	88.0

Result

Thus, the Pandas program was successfully executed to **find and replace missing values** in the given DataFrame. The missing values were replaced with **0**.

